Learning the Country, Not the Camera: Measuring and Explaining Contributor Shortcuts in CNN-Based Country Recognition

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Research question and design

Can a convolutional network classify the country of a street-level photo from geographic cues, or does it instead recognise the *photographer*? Mapillary imagery in OpenStreetView-5M (OSV-5M) [1] is heavily contributed: most photos of a country come from few recurring contributors whose cameras, mounts and colour profiles are persistent identifiers [9], a textbook shortcut [5, 11, 12]. Our question is quantitative: *how much of the accuracy of a country classifier is contributor identity, and where in the pipeline does it enter?* We study it with ResNet-18 [6] on 32 countries (those with $\geq$200 contributors), using a contributor-disjoint protocol: contributors are split 70/10/20 into train/val/test (stratified by main country), budgets per country are filled by water-filling with a per-contributor cap, and two additional evaluation sets hold out entire trips (linked by sequence or a $\sim$11 km cell) of *training* contributors. This yields *test* (12,800 images, 3,043 unseen contributors) and *test*$_{shared}$ (12,559 images, 3,037 seen contributors; $\leq$400 images per country each), over a training pool of 101,578 images from 11,132 contributors [7, 10]. Images are centre-cropped to 4:3 (removing the 16:9/4:3 camera cue) and stored at 128×170. We report balanced accuracy $\mathrm{BA} = \frac{1}{K}\sum_k R_k$ (mean per-class recall; chance $1/K = 3.1\%$) with 95% CIs from a bootstrap that resamples *contributors*, not images [3]; the statistic of interest is the contributor gap $\Delta = \mathrm{BA}(\mathrm{test}_{shared}) - \mathrm{BA}(\mathrm{test})$, computed on paired bootstrap draws so its interval is for the difference itself.

Models compared

All models share the same data, splits and input resolution, forming an ablation ladder over *how much of the network is fitted to the contributors*: **(i) Linear probe.** Frozen ImageNet [2, 8] ResNet-18 features (512-d global average pool) with L_2-penalised softmax regression, $\min_{W,b} \frac{1}{n}\sum_i \mathrm{CE}(W^\top f_i + b, y_i) + \lambda\|W\|_F^2$, fitted by L-BFGS on 9,600 images (300 per country); λ chosen on *val* (unseen contributors). **(ii) Partial fine-tune** (CPU interim, no GPU access so far): the stem and residual stages 1–2 stay frozen at ImageNet weights; stages 3–4 and the head train for 24 epochs of 20,480 images ($\approx$4.7 passes) with SGD (Nesterov, lr 0.02 per 256 images scaled by batch, cosine decay after 0.5-epoch warm-up, weight decay 10^{-4} off BatchNorm/bias), dropout 0.2, label-smoothed cross-entropy ($\varepsilon = 0.1$: $q_y = 1 - \varepsilon + \varepsilon/K$, $q_{k\neq y} = \varepsilon/K$), random resized crops (scale 0.6–1), and *no* horizontal flip (mirroring swaps the driving side). **(iii) Staged data extension**: 261,572 additional images exist for our countries under a 100-image-per-contributor cap (contributor concentration again), dealt into three $\sim$87k chunks; each chunk is trained for five coverage epochs (every image exactly once) from the previous best checkpoint, with inverse-frequency class weights and a 1,600 replay floor per country, giving cumulative distinct-image counts of 189k/276k/363k. **(iv) k-nearest neighbours** on the same pooled features (z-scored on the gallery, cosine, $k \in \{1, 15\}$): a non-parametric alternative that also makes the shortcut directly observable in retrieval.

Table 1: Balanced accuracy (%) on the two test protocols with 95% contributor-clustered CIs (test$_{shared}$ point estimates; CIs omitted for width) and the median great-circle distance of *errors* to the predicted country's centroid. Paired bootstrap vs. the fine-tune: stage 3 test +2.0 [1.4, 2.5], gap +1.6 [0.6, 2.5].

model	test	test$_{sh.}$	Δ	err. km
probe (frozen)	13.5 [12.8,14.3]	13.7	+0.2 [−1.1, 1.4]	5543
fine-tune	28.7 [27.5,29.8]	34.5	+5.8 [4.1, 7.5]	2439
+ chunk 1	29.8 [28.6,31.0]	36.8	+7.0 [5.2, 8.8]	2679
+ chunk 2	30.1 [28.9,31.3]	37.7	+7.6 [5.8, 9.4]	2852
+ chunk 3	**30.7** [29.4,31.9]	**38.0**	+7.4 [5.6, 9.2]	2865
15-NN (frozen)	9.7†	11.5†	—	—
15-NN (stage 3)	25.3†	32.8†	—	—

†top-1; gallery val $\rightarrow$ test, val$_{shared}$ $\rightarrow$test$_{shared}$.

Results: scaling buys little and widens the gap

Table 1 and Fig. 1a give the headline. Fitting the upper network *doubles* accuracy over frozen features (13.5$\rightarrow$28.7% BA). Tripling the data again adds only +2.0 pp on unseen contributors while contributors already seen gain +3.5 pp: Δ widens from +5.8 to +7.4 pp. Because new images are mostly *more trips of the same contributors*, scale adds contributor identity faster than geographic knowledge — consistent with the per-epoch curve, where the val gap $\mathrm{BA}(\mathrm{val}_{shared}) - \mathrm{BA}(\mathrm{val})$ stays near zero below $\sim$37% training accuracy (epochs 0–7) and grows to +3–4 pp once the network fits the training set ($>$ 50%). Errors are not random either: confusions stay local (Spain$\rightarrow$Italy, Czechia$\rightarrow$Poland), the worst classes are western-European (France 5%), and the median misclassification lands 5,543 km from the truth for the probe but 2,439 km after fine-tuning (correct answers sit $\sim$220 km from their centroid): fine-tuning mostly removes continent-level mistakes, and lifts top-3 test accuracy from 29.4 to 52.4%. On frozen features a 15-NN reaches only 9.7% top-1 (its trained head: 13.5%); after fine-tuning the gap narrows to 25.3 vs 30.7%, so fine-tuning chiefly makes the features *metrically* country-ordered.

Anatomy of the shortcut: three convergent analyses

(1) Exposure dose–response. If Δ were geography (seen contributors photographing typical places), accuracy on their held-out trips would be flat in their training exposure. Instead, we bin each test$_{shared}$ contributor by training images and compute the country-adjusted excess $E_B =$

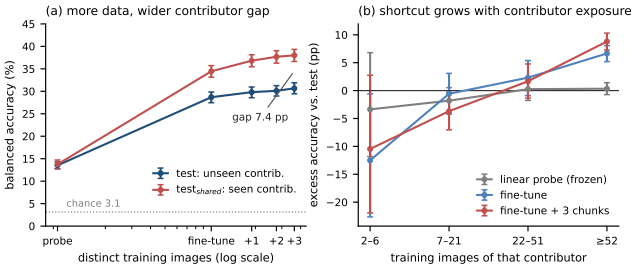


Figure 1: (a) Balanced accuracy versus distinct training images: the two protocols diverge as data grows. (b) Country-adjusted excess accuracy on test$_{shared}$ (held-out trips of seen contributors), binned by the contributor's number of training images, relative to the same countries' accuracy on the disjoint test split; whiskers are 95% contributor-bootstrap CIs. Frozen features show no trend; fine-tuned models gain up to +8.8 pp for the most-exposed contributors.

$\frac{1}{|B|}\sum_{i\in B}\left(\mathbf{1}[\hat{y}_i = y_i] - a_{c(i)}\right)$, where a_c is the same model's accuracy on country c in the disjoint test split. The probe is flat (+0.3 pp, CI $[-0.7, +1.4]$, for contributors with ≥52 images), whereas stage 3 gives +1.7 $[-1.3, +4.8]$ at 22–51 images and +8.8 pp $[+7.3, +10.3]$ at ≥52 images (median exposure 99; 84% of images; Fig. 1b). The advantage is a *memorisation dose–response*, not a sampling artefact. **(2) Retrieval view.** In the 1-NN classifier (val$_{shared}$ gallery), 6.9% of test$_{shared}$ queries retrieve a photo by the *same contributor* as nearest neighbour; those queries are 90.6% correct, the others 22.0%. This single mechanism contributes $0.069 \times (0.906 - 0.220) \approx +4.7$ pp of top-1 — most of the +7.0 pp top-1 gap (37.6% vs 30.7%) — and the comparison is within the same query set, so it is paired. The same holds on frozen features (5.0% of queries, then 89.6% vs 7.0% correct): the retrieval cue is available to any photographer-clustered representation; only the *fitted classifier* turns it into an accuracy gap. **(3) Feature-space verification.** We score pairs of *test* images (contributors never seen in training) by cosine similarity: same-photographer pairs from different sequences versus different-photographer pairs from the same country, so country information cannot help; AUC is 0.5 at chance. Frozen ImageNet features already separate photographers (AUC 0.59); fine-tuned features score 0.57, with a 14% nearest-neighbour same-contributor rate against a 0.1% random-pair chance. Photographer style is thus present in the representation regardless of training — yet the probe, built on the *more* photographer-decodable features, shows no gap and no exposure trend (Fig. 1b). Decodable style alone does not create the shortcut; it appears only when the classifier is fitted to specific contributors (analyses 1–2).

Problems encountered and decisions

- *No GPU access yet.* Kaggle GPU quotas remain unconfigured, so we redesigned the plan rather than the question: freeze stem and stages 1–2 (17 vs 10 img/s), subsample epochs to 20,480 images, make training resumable (checkpointed RNG and data order), and train the data extension chunk-by-chunk. The GPU protocol (below) is unchanged and ready.
- *Static-pixel camera masks failed.* Temporally-static pixels should mark bonnets and dashboards, but no pixel had across-photo std < 15 for the 60 largest contributors, and the few stable regions were roads, snow and sky — country cues whose shapes would themselves be fingerprints. Masks stay opt-in and frame-capped; instead we blank the bottom 20% of every image, re-encode JPEG (quality 30–95) and jitter white balance/brightness/contrast/saturation. Hue is deliberately *not* jittered (road-marking colour is a country cue); blanking—not cropping—keeps geometry identical. This pipeline is implemented but awaits the GPU runs.
- *Warm restarts.* Stage 1 restarted at lr 0.01 and lost 3 pp of val BA before recovering; stages 2–3 restart at 0.004 with no dip. Stage 1 also introduced inverse-frequency class weights together with new data, so that step does not cleanly separate data from objective; the within-stage trajectories and the exposure analysis (which is unaffected) carry the interpretation.
- *Data engineering.* 139,696 single JPEGs read out of 98 remote OSV-5M zips by HTTP range requests (42 min) instead of a 245 GB download.

Plan to completion

On GPU (Kaggle T4): the 2 × 2 design — {ImageNet, from-scratch} × {standard, fingerprint-removal} — testing whether bottom-band blanking and photometric jitter close Δ and at what cost in disjoint-test accuracy; end-to-end fine-tuning; a wider λ grid for the probe (its optimum sits at the grid edge); Grad-CAM qualitative analysis; and possibly a domain-adversarial contributor penalty [4]. The final report adds per-class and confusion analysis and the interactive "Dashcam Duel" demonstration.

Data: OSV-5M, CC-BY-SA-4.0 [1]; code and scripts in the project repository. AI use: ZCode (GLM) assisted with pipeline/analysis code and produced a first draft of this text, which the authors revised, verified claim by claim against the run artifacts, and take full responsibility for.

References

[1] Guillaume Astruc, Nicolas Dufour, Ioannis Siglidis, Constantin Aronssohn, Nacim Bouia, Stephanie Fu, Romain Loiseau, Van Nguyen Nguyen, Charles Raude, Elliot Vincent, Lintao Xu, Hongyu Zhou, and Loic Landrieu. OpenStreetView-5M: The many roads to global visual geolocation. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 21967–21977, 2024. doi: 10.1109/CVPR52733.2024.02074.

[2] Jia Deng, Wei Dong, Richard Socher, Li-Jia Li, Kai Li, and Li Fei-Fei. ImageNet: A large-scale hierarchical image database. In *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 248–255, 2009.

[3] Christopher A. Field and Alan H. Welsh. Bootstrapping clustered data. *Journal of the Royal Statistical Society: Series B (Statistical Methodology)*, 69(3):369–390, 2007.

[4] Yaroslav Ganin, Evgeniya Ustinova, Hana Ajakan, Pascal Germain, Hugo Larochelle, François Laviolette, Mario Marchand, and Victor Lempitsky. Domain-adversarial training of neural networks. *Journal of Machine Learning Research*, 17(59):1–35, 2016.

[5] Robert Geirhos, Jörn-Henrik Jacobsen, Claudio Michaelis, Richard Zemel, Wieland Brendel, Matthias Bethge, and Felix A. Wichmann. Shortcut learning in deep neural networks. *Nature Machine Intelligence*, 2(11):665–673, 2020.

[6] Kaiming He, Xiangyu Zhang, Shaoqing Ren, and Jian Sun. Deep residual learning for image recognition. In *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 770–778, 2016.

[7] Shachar Kaufman, Saharon Rosset, Claudia Perlich, and Ori Stitelman. Leakage in data mining: Formulation, detection, and avoidance. *ACM Transactions on Knowledge Discovery from Data*, 6(4):15:1–15:21, 2012.

[8] Simon Kornblith, Jonathon Shlens, and Quoc V. Le. Do better ImageNet models transfer better? In *Proceedings of the*

IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), pages 2661–2671, 2019.

[9] Jan Lukáš, Jessica Fridrich, and Miroslav Goljan. Digital camera identification from sensor pattern noise. *IEEE Transactions on Information Forensics and Security*, 1(2):205–214, 2006.

[10] David R. Roberts, Volker Bahn, Simone Ciuti, Mark S. Boyce, Jane Elith, Gurutzeta Guillera-Arroita, Severin Hauenstein, José J. Lahoz-Monfort, Boris Schröder, Wilfried Thuiller, David I. Warton, Brendan A. Wintle, Florian Hartig, and Carsten F. Dormann. Cross-validation strategies for data with temporal, spatial, hierarchical, or phylogenetic structure. *Ecography*, 40(8):913–929, 2017.

[11] Antonio Torralba and Alexei A. Efros. Unbiased look at dataset bias. In *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 1521–1528, 2011.

[12] John R. Zech, Marcus A. Badgeley, Manway Liu, Anthony B. Costa, Joseph J. Titano, and Eric Karl Oermann. Variable generalization performance of a deep learning model to detect pneumonia in chest radiographs: A cross-sectional study. *PLoS Medicine*, 15(11):e1002683, 2018.